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C01 - ASSESSMENT TOOL 1 -Application Based Questions

Course Code: DSA03

Course Title: Natural Language Processing

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1. Resume Information Extraction System Using Regular Expressions

The Resume Information Extraction System is designed to automate the process of analyzing resumes received by a recruitment company. Manual resume screening requires considerable time and effort, especially when dealing with thousands of applications. To solve this problem, Python's built-in re module is used to extract important information from resume text using predefined Regular Expression patterns.

The system extracts the candidate's name, email address, mobile number, technical skills, and years of experience. It identifies skills such as Python, Java, SQL, Machine Learning, and Natural Language Processing (NLP). The extracted information is organized into a structured profile containing all the candidate's details.

After extracting the information, the system checks the candidate's eligibility based on two conditions: the candidate must have at least two years of experience and must possess the Python skill. Candidates who satisfy both conditions are shortlisted, while others are marked as ineligible. Finally, the system displays the candidate profiles and a list of eligible applicants.

Result: The system successfully automates resume information extraction, generates structured candidate summaries, and shortlists eligible candidates, thereby reducing manual screening effort and improving recruitment efficiency.

Candidate Profile
Name: Rahul Sharma Email: rahul.sharma@gmail.com Mobile: 9876543210 Skills: ['Python', 'Java', 'SQL', 'Machine Learning'] Experience: 3 Eligible: Yes
Candidate Profile
Name: Priya Kumar

Email: priya.kumar@gmail.com Mobile: 9123456780 Skills: ['Java', 'SQL', 'NLP'] Experience: 1 Eligible: No
Candidate Profile
Name: Arjun Reddy Email: arjun.reddy@gmail.com Mobile: 9988776655 Skills: ['Python', 'SQL', 'Machine Learning', 'NLP'] Experience: 5 Eligible: Yes
Candidate Profile
Name: Sneha Patel Email: sneha.patel@gmail.com Mobile: 9012345678 Skills: ['Python', 'Java'] Experience: 2 Eligible: Yes
ELIGIBLE CANDIDATES
Name: Rahul Sharma, Experience: 3 years, Skills: Python, Java, SQL, Machine Learning
Name: Arjun Reddy, Experience: 5 years, Skills: Python, SQL, Machine Learning, NLP
Name: Sneha Patel, Experience: 2 years, Skills: Python, Java

2. Product Search System Using Regular Expressions

The Product Search System is developed to improve product discovery on an e-commerce platform. Customers may search for products using complete names, the beginning or ending of a product name, or only a portion of a keyword. To support these requirements, Python's built-in re module is used to perform different types of searches on a predefined list of product names.

The system supports five search methods: exact keyword search, prefix-based search, suffix-based search, partial keyword search, and case-insensitive search. Exact keyword

search identifies a complete product name that matches the search term. Prefix-based search finds products whose names begin with the given keyword, while suffix-based search finds products whose names end with it. Partial keyword search identifies products containing the keyword anywhere in their names. Case-insensitive search allows users to find products regardless of whether they enter uppercase or lowercase letters.

After each search operation, the system displays all matching product names. It also calculates the total number of matching products and generates a search report. Regular Expression functions such as `re.fullmatch()`, `re.match()`, and `re.search()` are used to implement the different matching techniques.

Result: The system successfully performs flexible product searches, displays all matching products, and generates a report containing the number of matches for each search. This improves product accessibility and provides a more convenient search experience for e-commerce customers.

Search Type: EXACT
Keyword: Apple iPhone 15 Matching Products: - Apple iPhone 15 Total Matching Products: 1
Search Type: PREFIX
Keyword: Apple Matching Products: - Apple iPhone 15 - Apple MacBook Air - Apple Watch Total Matching Products: 3
Search Type: SUFFIX
Keyword: Laptop Matching Products: - HP Laptop - Dell Laptop Total Matching Products: 2
Search Type: PARTIAL
Keyword: Galaxy Matching Products: - Samsung Galaxy S24 - Samsung Galaxy Buds Total Matching Products: 2

3. University Registration System Using Regular Expressions

The University Registration System is developed to validate student information entered through an online registration portal. Accurate student details are essential for maintaining university records and ensuring that students enroll in the correct courses. Manual verification can be time-consuming and may result in data-entry errors. Therefore, the system uses Python's built-in `re` module to validate student details according to predefined formatting rules.

The system validates five important fields: student register number, institutional email address, course code, semester information, and mobile number. Each field is checked using a specific Regular Expression pattern. The register number is verified according to the university's prescribed format, while the institutional email is checked for a valid structure and the approved domain. The course code is validated according to its required combination of letters and digits. The semester is checked to ensure that it falls within the permitted range, and the mobile number is verified to ensure that it follows the accepted format.

For every field, the system displays an appropriate validation message indicating whether the entered information is valid or invalid. The validation results are stored and used to determine the final registration status. If all five fields pass validation, the registration is marked as successful. If any field is invalid, the registration is marked as failed, and the student is asked to correct the invalid information.

Result: The system successfully validates student information, provides field-wise validation messages, and generates a final registration status report. It helps reduce data-entry errors, improves the accuracy of student records, and makes the university registration process more reliable and efficient.

VALIDATION REPORT
Register Number: Valid Institutional Email: Valid Course Code: Valid Semester: Valid Mobile Number: Valid
FINAL REGISTRATION STATUS
Registration Successful! All student details are valid.